

EVOLUTION OF DIGITAL ELECTRONICS

Reverse Engineering a Modern Chip

1. Question.



What is inside a smartphone, laptop, car ECU, or AI server?



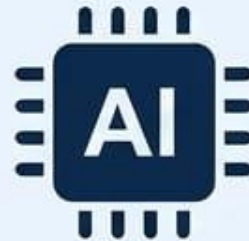
Phone



Laptop

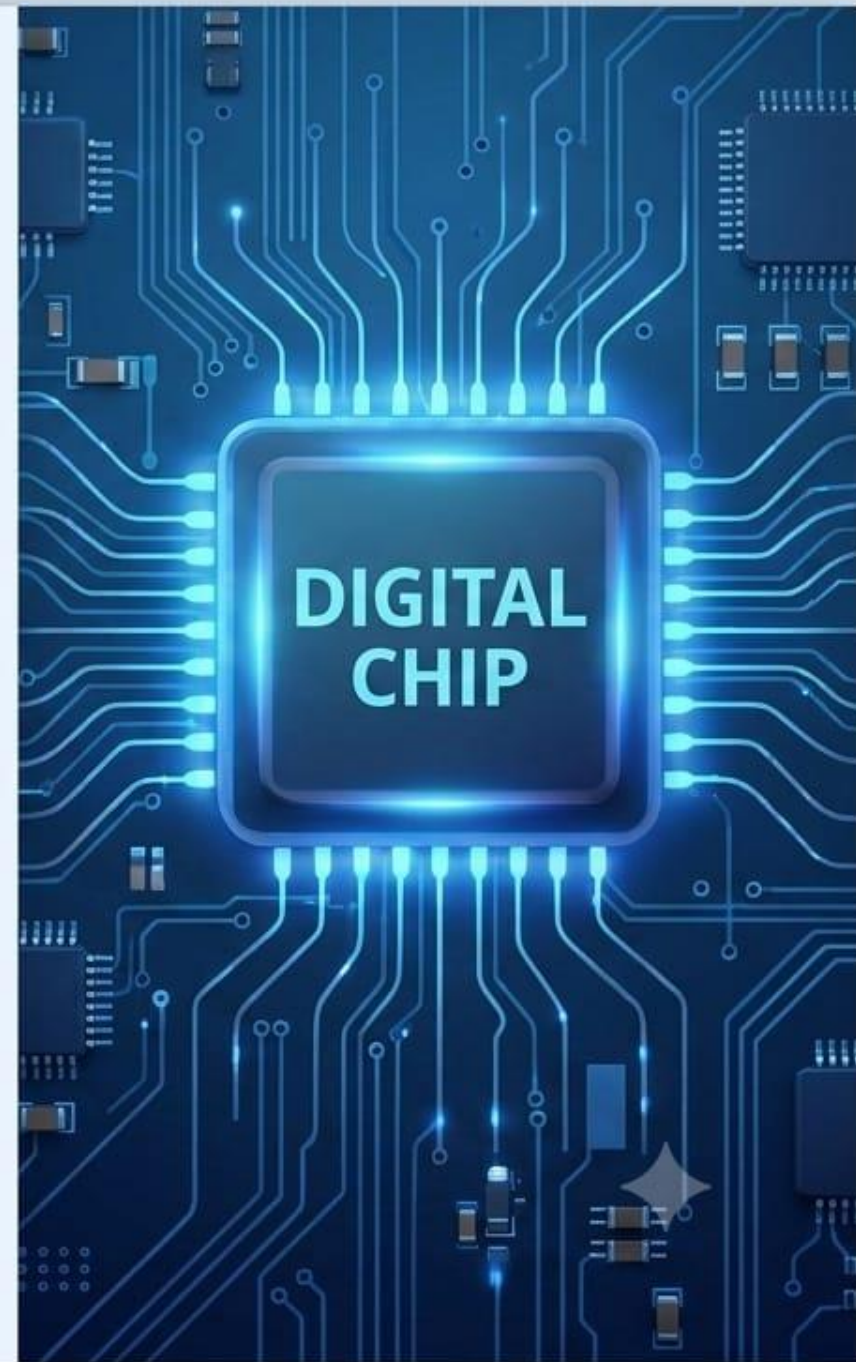


Car ECU



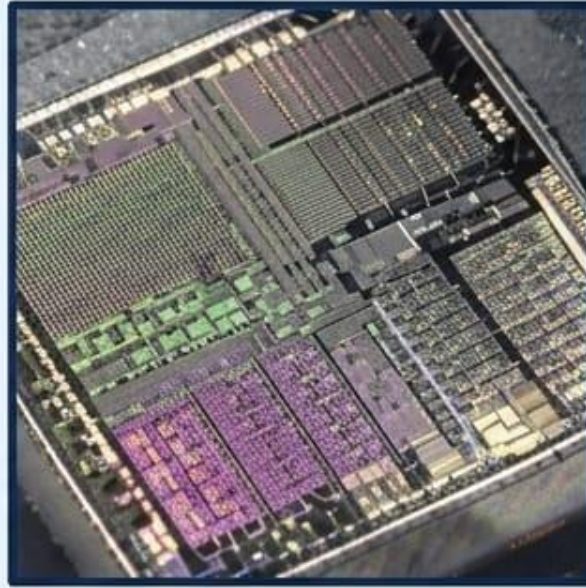
AI

Answer: A digital chip



2. Inside a Modern Chip

- CPU
- Memory
- Cache
- Interconnects
- I/O controllers
- Billions of transistors



Building Blocks



Result

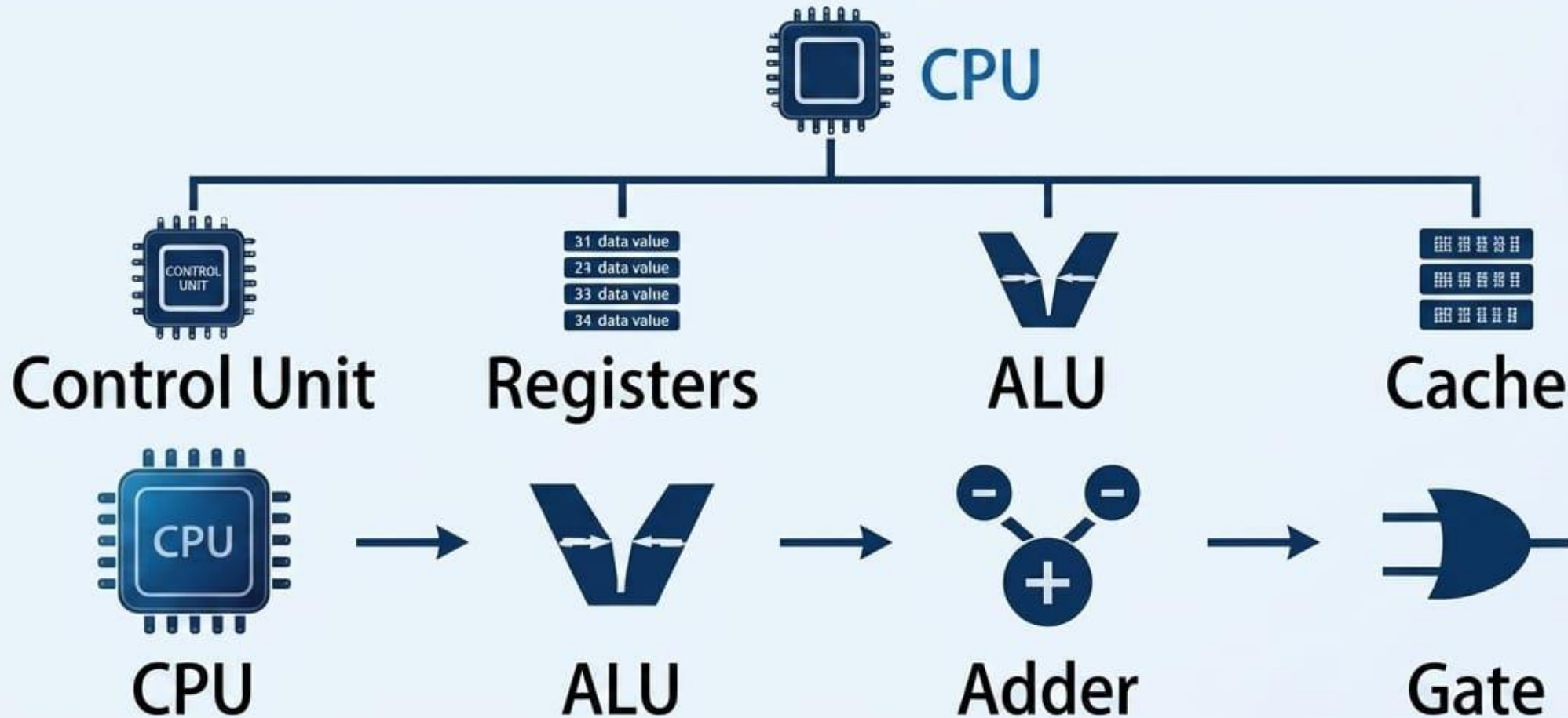
Concept

Building Block

Result

3. CPU Breakdown

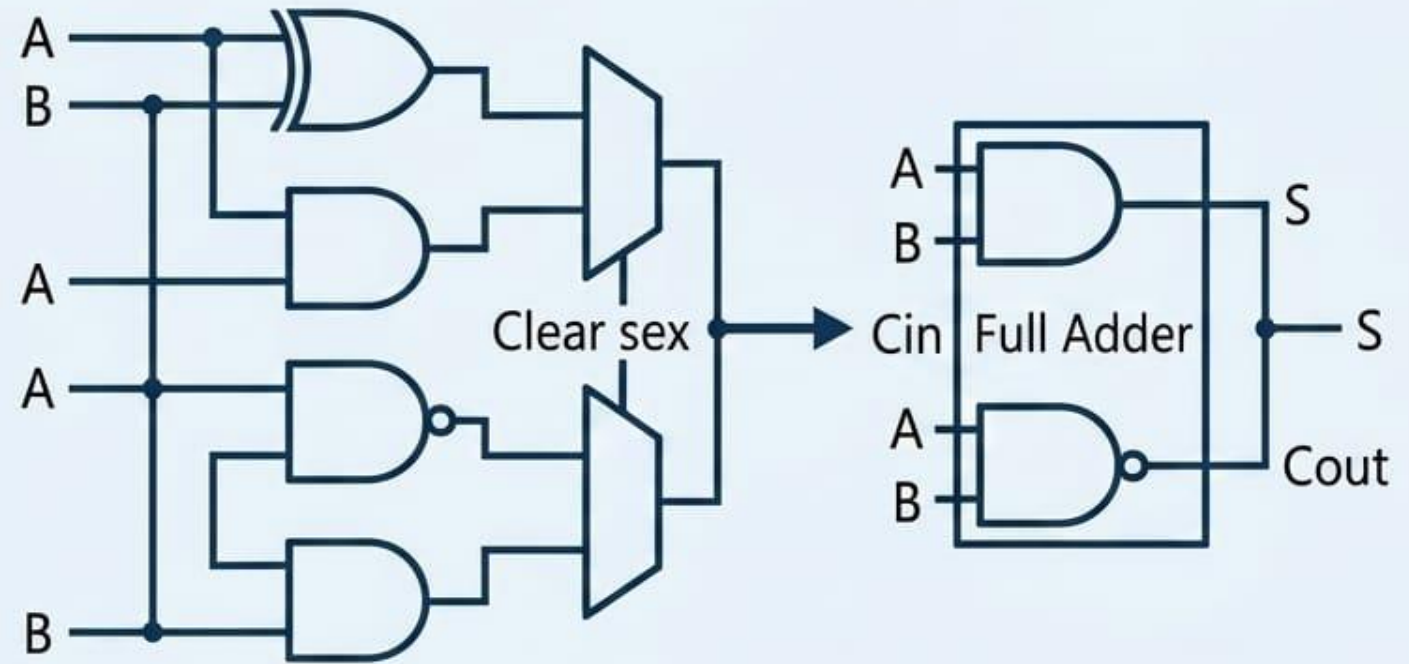
CPU contains Control Unit, Registers, ALU and Cache.



4. Inside the ALU

Arithmetic and Logic operations are implemented using...

- Half Adders
- Full Adders
- Multiplexers
- Decoders
- Encoders
- Comparators



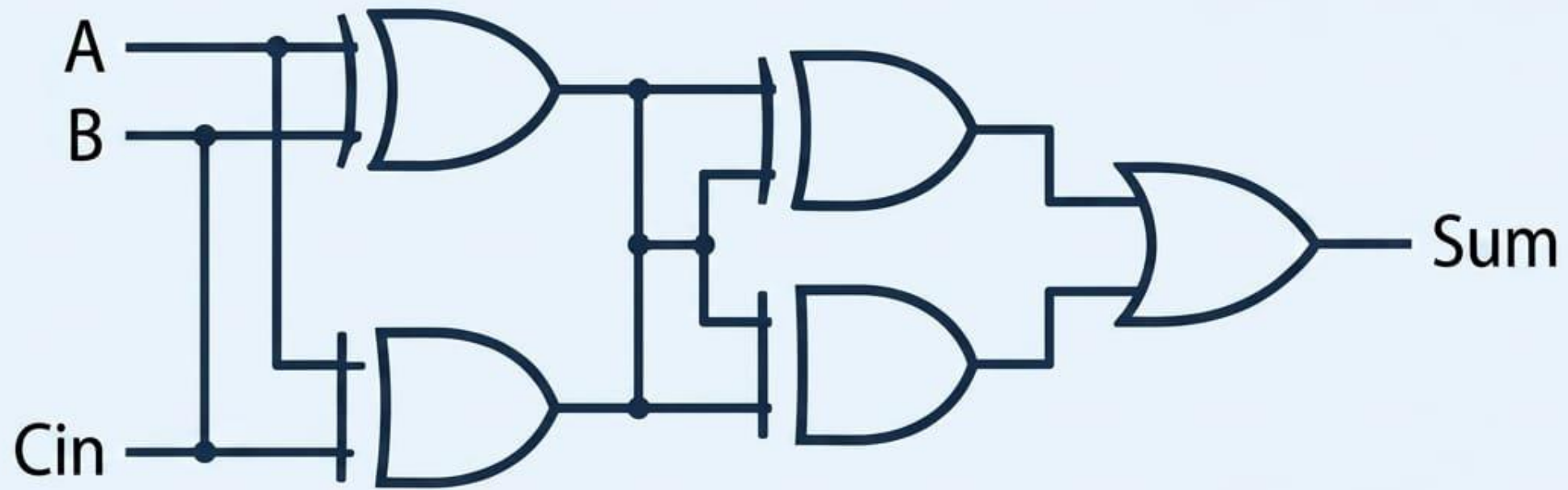
Concept

Building Block

Result

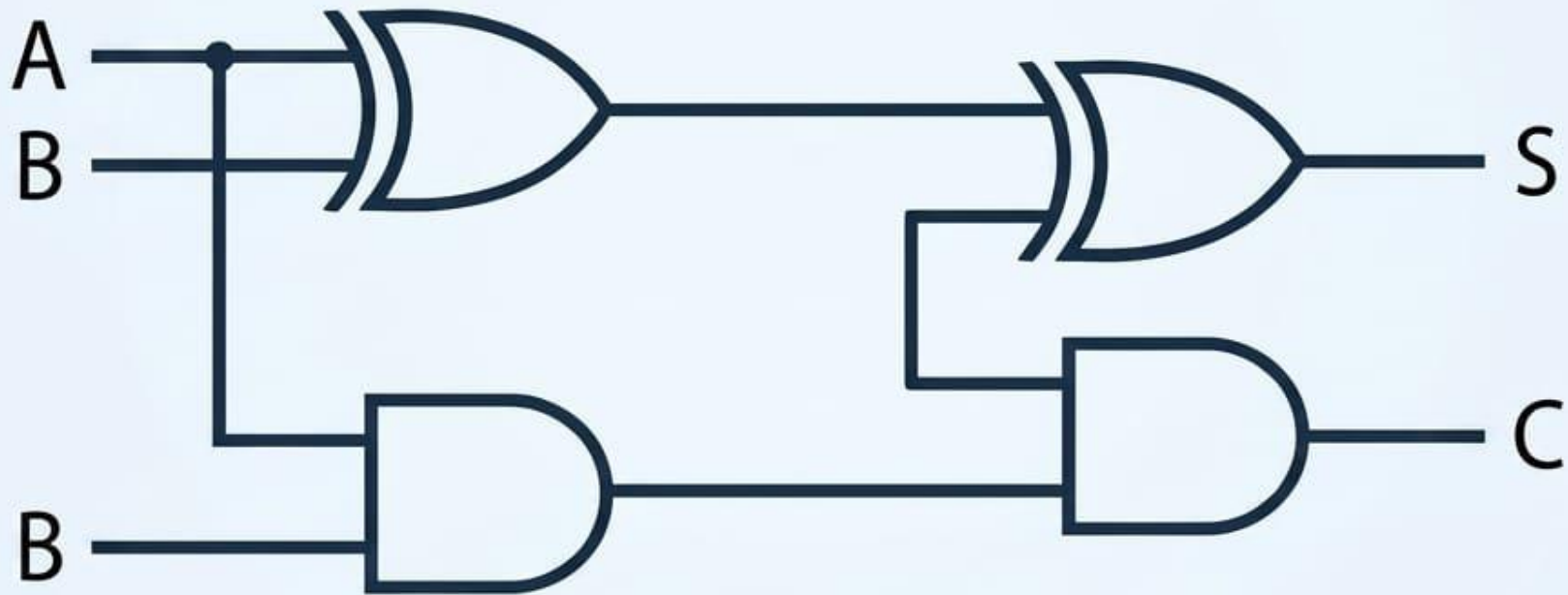
5. Full Adder

A Full Adder can be built from two Half Adders and one OR gate.



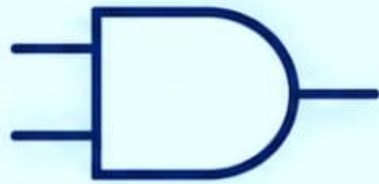
6. Half Adder

A Half Adder consists of XOR and AND gates.

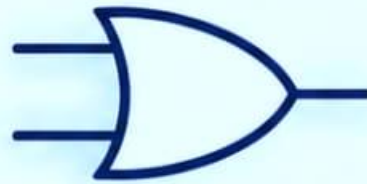


7. Logic Gates

AND, OR, NOT, NAND, NOR, XOR, XNOR are the foundation of digital logic.



AND



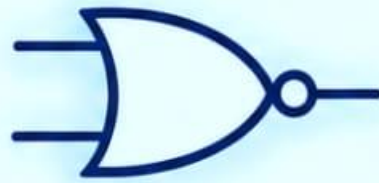
OR



NOT



NAND



NOR



XOR



XNOR

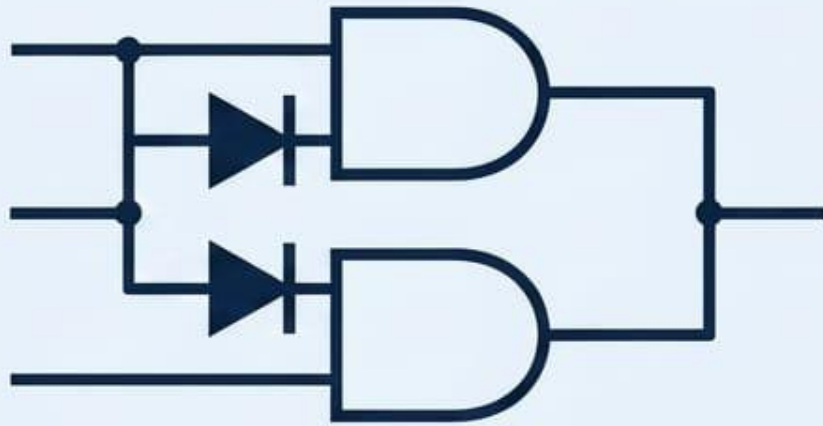
Concept

Building Block

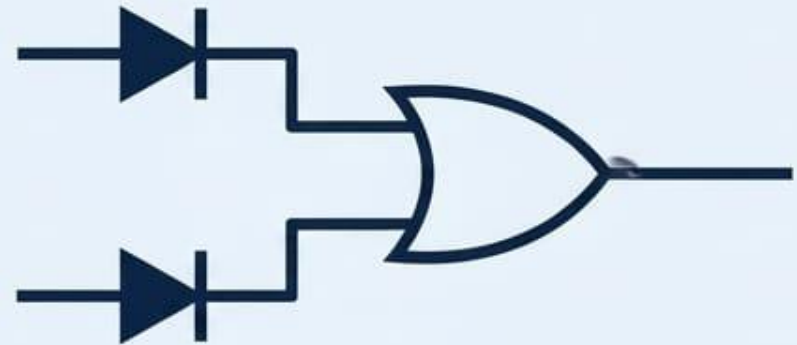
Result

8. Diode Logic

Diodes can implement basic AND and OR logic by controlling current flow.



Diode AND



Diode OR

Concept

Building Block

Result

9. Why Transistors?

Diodes cannot amplify or restore logic levels.
Transistors solved these limitations.



Diode

VS



Transistor

Amplification & Switching

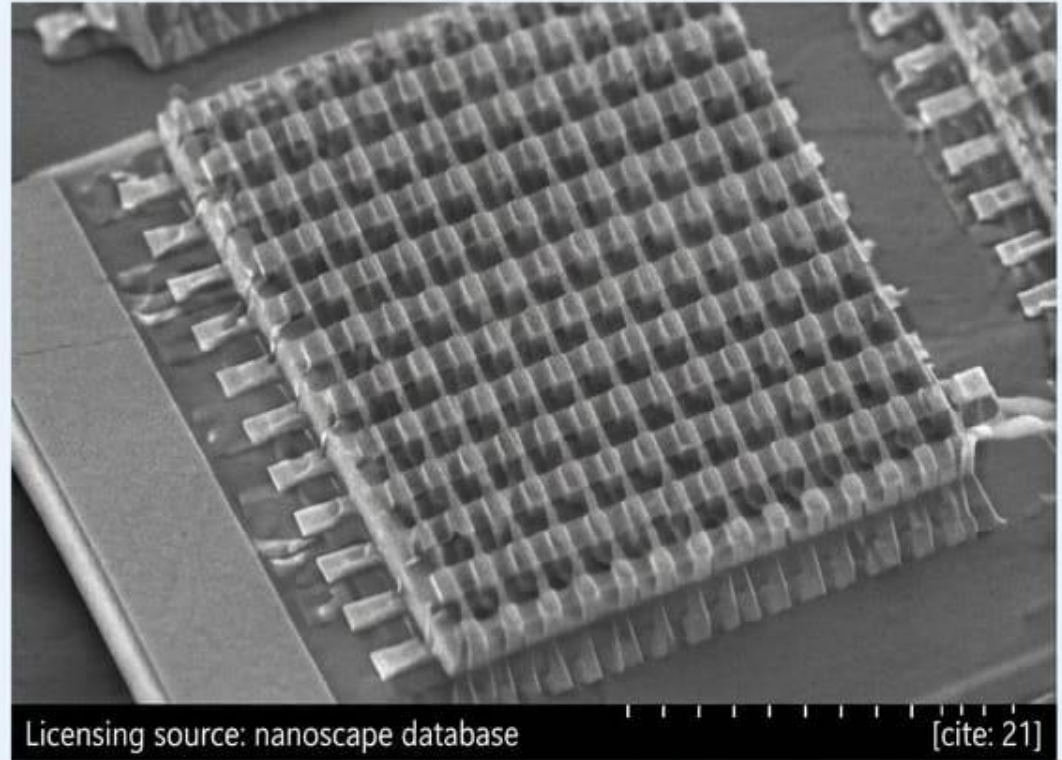
Concept

Building Block

Result

10. Modern CMOS

Modern processors use billions of MOSFET transistors to build logic gates.



Licensing source: nanoscape database

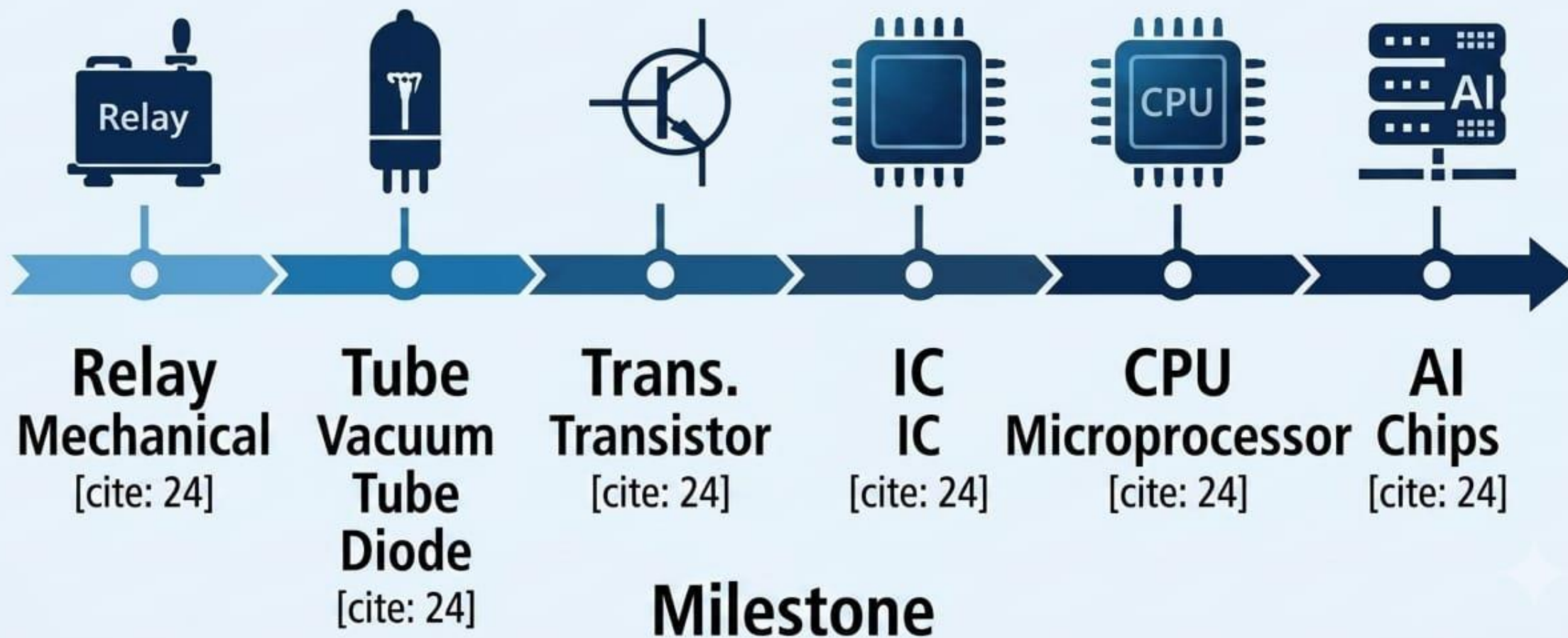
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Concept

Building Block

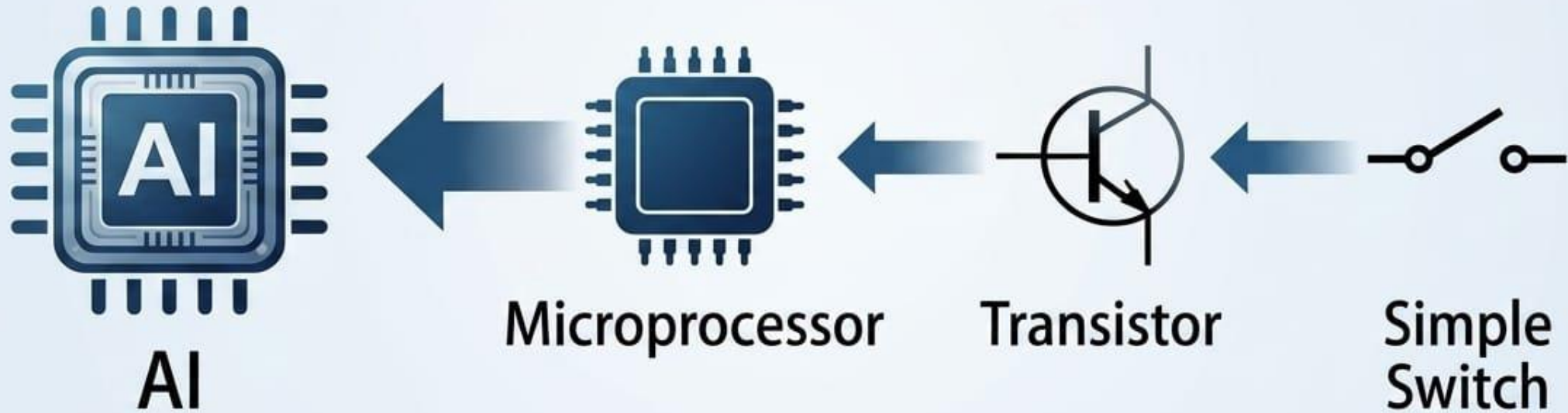
Result

11. Evolution Timeline



12. Summary

Every modern AI processor ultimately traces back to simple electronic switching principles.



Concept

Building Block

Result